

Olivia Griffith

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EXPERIENCE

Recent Astrophysics Graduate from Michigan State University's Lyman Briggs College with research experience in astronomy and physics, specializing in computational modeling. Skilled in analytical problem-solving, driven by a strong work ethic, and actively engaged in extracurricular activities. Passionate about advancing knowledge and contributing to innovative research through a career in space sciences or similar STEM fields.

SKILLS

- *Programming: Python, Jupyter, Anaconda, Mathematica, SQL, Data Analysis, Data Reduction*
- *Image Analysis: CASA, DS9, AstrolmageJ, N.I.N.A, Photometry, Light-Curve Analysis, Astropy*
- *Soft Skills: Microsoft Office Suite, Communication, Time management, Scientific Writing, Strong Interpersonal Skills*
- *Leadership: Organizational Skills, Teaching Experience, Project Management, Lesson Planning, Adaptability*

EDUCATION

Michigan State University – East Lansing, MI

GPA: 3.87 on a 4.0 scale - Dean's List all semesters

Bachelor of Science, Astrophysics

September 2021 - May 2025

Computational Mathematics, Science, and Engineering Minor

Lyman Briggs College

- Thermodynamics and Modern Physics Advisory Committee
- Department of Physics and Astronomy - Office Assistant

EXPERIENCE

Michigan State University - East Lansing, MI

Research Assistant - Funded by NSF grants AST-2107070 and AST-2205628.

May 2024 - Present

- Worked collaboratively with postdoctoral researchers to investigate progenitors of Type Ia-CSM supernovae through modeling rates of mass-loss obtained via radio continuum observations.
- Spearheaded computational modeling efforts and contributed extensively to the research paper, serving as the corresponding author.
- Submitted for publication, undergoing referee report revisions.
- Presented research at the Mid-Michigan Symposium for Undergraduate Research Experiences (MID-SURE).

May 2025 - Present

- Processed archival NRAO supernova data through CASA.
- Tasks include gathering data based on predetermined parameters, imaging and flagging data, and data calibration and analysis.

MSU Observatory Research Program - East Lansing, MI

Intermediate Observer

September 2023 - May 2025

- Managed telescope and dome operations, target tracking, and camera adjustments during overnight observations of various astronomical targets including white dwarfs and exoplanets.
- Utilized the N.I.N.A. application to collect diverse datasets and applied advanced image analysis techniques using Jupyter Notebooks and AstrolmageJ, processing hundreds of images to extract key astronomical insights.

Cranbrook Schools - Bloomfield Hills, MI

STEM Instructor

May 2023 - August 2023

- Delivered engaging, hands-on STEM lessons using MIT-developed modules on topics including physics, biology, rocketry, computing and 3D printing.
- Developed and adapted lesson plans appropriate for student age and ability levels.
- Oversaw camper activities, prioritizing safety and engagement, and fostering positive, enriching interactions.

INVOLVEMENT

- *MSU Astronomy Club - Member*
- *United States Polo Association - Riding Member*
- *United States Pony Club - Member*
- *MSU Polo Club - Vice President, Varsity Member*
- *MSU Casual Music Club - Multi-Instrumentalist*
- *MSU Wakeboarding Club - Member*
- *Stanford Physics, Identity, and Equity Program (PIE)*
- *MSU Stellar Mentorship Program*
- *Women and Minorities in the Physical Sciences (WaMPS @ MSU)*

ACCOLADES

- *Lawrence W. Hantel Endowed Fellowship Fund, in Memory of Professor Donald J. Montgomery*
- *Outstanding Senior Award - Department of Physics and Astronomy*
- *Best LB Chemistry II Symposium Presentation 2022*
- *Best LB Biology I Group Proposal Paper 2022*